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(54) **COOLER BAG**

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(57) **ABSTRACT**

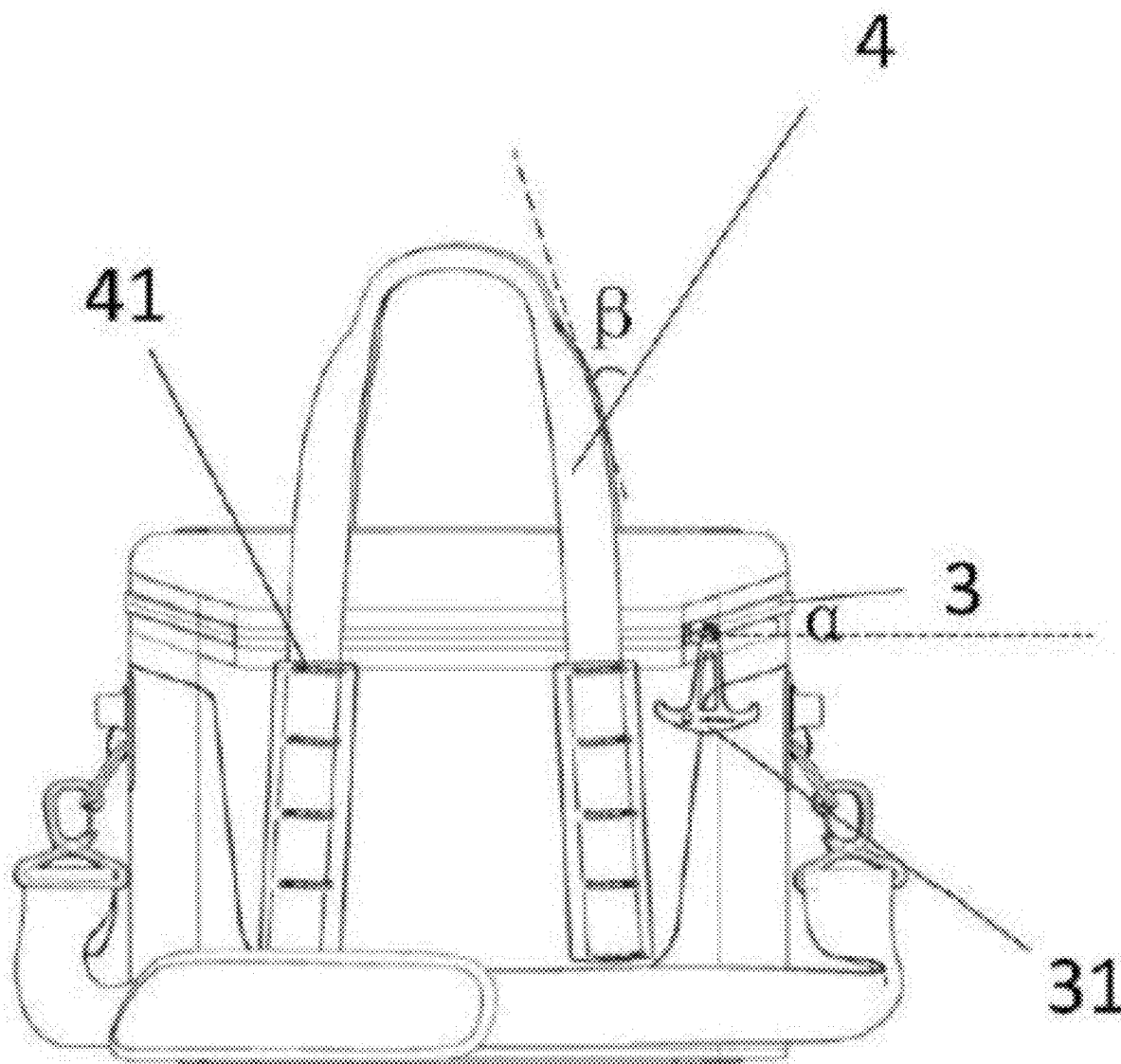
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A cooler bag, comprising: a box body (2) and a cover body (1). The box body (2) and the cover body (1) are connected by means of a zipper (3). The zipper (3) is provided with a pull tab (31). A portion of the pull tab (31) in contact with fingers of a user is arc-shaped.



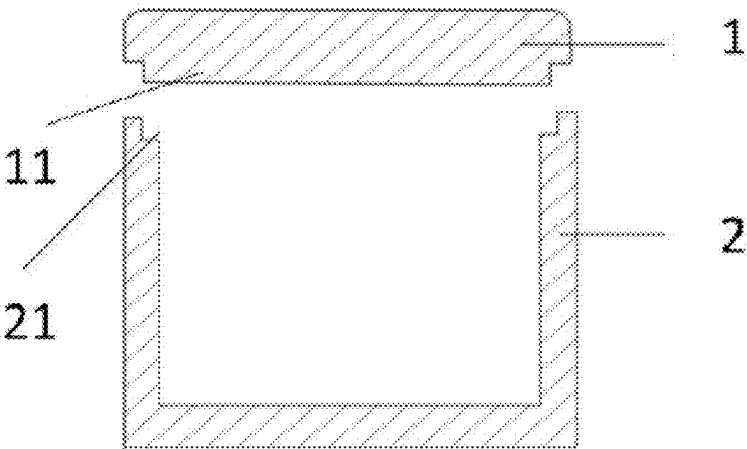


FIG.3

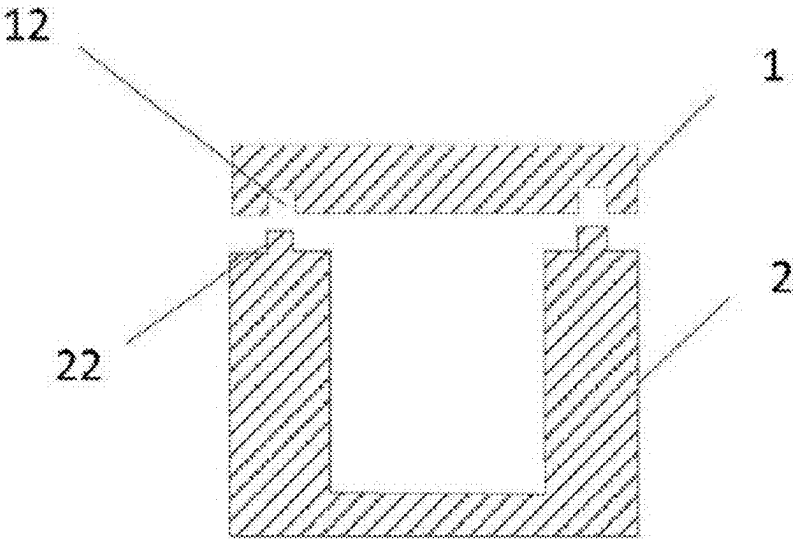


FIG.4

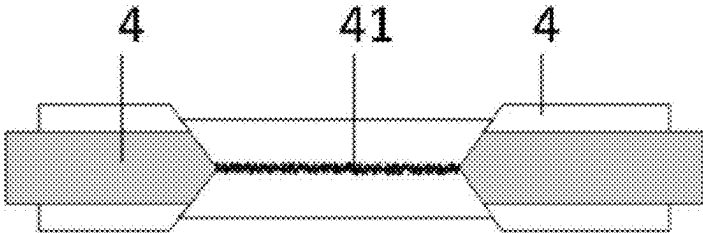


FIG.5

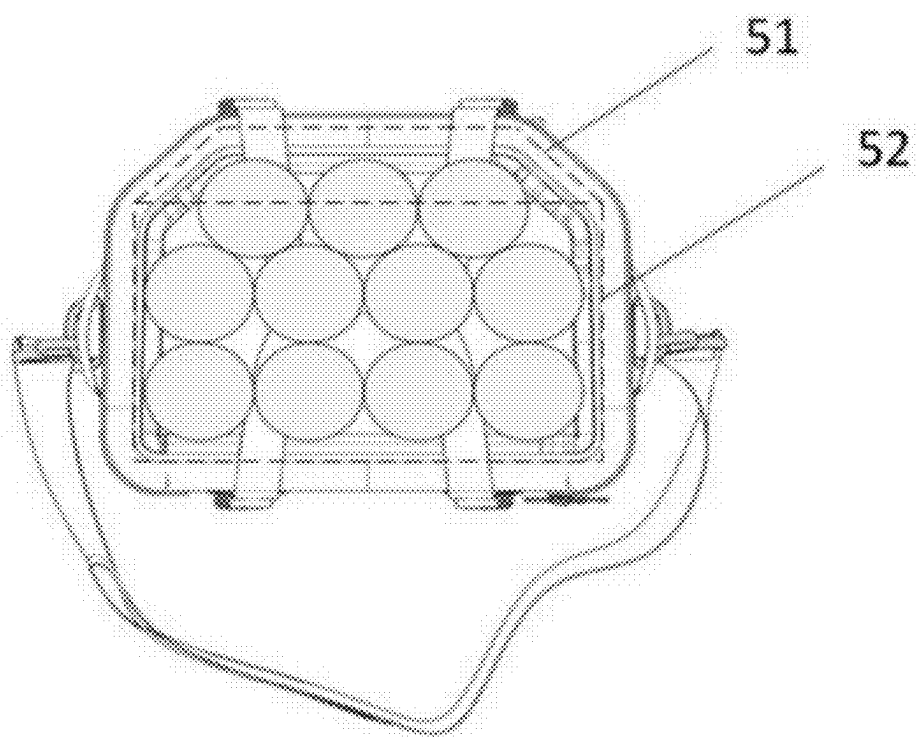


FIG.6

COOLER BAG

BACKGROUND OF THE INVENTION

[0001] The present utility model relates to a power outlet, and more especially, to a magic square power outlet which can be assembled and connected.

[0002] In the prior art, to guarantee the airtightness of ice bags, the zipper of the ice bags is designed to be tight and can be provided with only one zipper pull; if two zipper pulls are adopted, a seam will be formed at the joint of the two zipper pulls, which will compromise the airtightness and thermal-insulation property of the ice bags. An existing solution to this problem in two aspects: in one aspect, the zipper pull in contact with the fingers of users is designed to be T-shaped or circular, such that force can be easily applied with the fingers; in the other aspect, other mechanical parts or plastic parts rather than zippers are used to open or close the ice bags, such that the ice bags can be opened only by opening the mechanical parts. In addition, in the prior art, thermal-insulation cotton in the ice bags is used for thermal insulation; however, a blind area not covered with the thermal-insulation cotton exists at the position where the zipper is located. An existing solution to this problem is to thicken cotton at the top, such that the cotton can exactly fit cotton on the bag body after the ice bags are closed, thus reducing the seam between the zipper and the bag body. Moreover, the ice bags in the prior art are heavy due to their material properties, so the overall weight of the ice bags will be very large when the ice bags are used for containing drinks, food, and the like. An existing solution to this problem is to reinforce the force-bearing portion with bar-tack.

[0003] Thus, the prior art has the following technical problems:

[0004] 1. The zipper for opening or closing voltage-type ice bags is difficult to pull.

[0005] 2. Heat may leak from the zipper of the voltage-type ice bags, thus compromising the thermal-insulation property.

[0006] 3. The force-bearing capacity of the handle of the voltage-type ice bags is poor.

[0007] 4. The utilization rate of the internal space of the voltage-type ice bags is low.

BRIEF SUMMARY OF THE INVENTION

[0008] The main objective of the application is to solve at least one of the above-mentioned technical problems. To solve at least part of the above-mentioned technical programs, the application provides an ice bag, comprising a box and a cover;

[0009] The box and the cover are connected through a zipper which is provided with a zipper pull, and a portion, in contact with the fingers of users, of the zipper pull is arc.

[0010] Optionally, when the ice bag is placed flat, a non-zero first angle is formed between at least a tail of the zipper and a horizontal direction.

[0011] Optionally, a protruding portion is disposed on a lower surface of the cover, and a concave portion is disposed on an upper surface of the box; and when the cover covers the box, the protruding portion is inlaid in the concave portion, such that the internal space of the box is sealed by the cover.

[0012] Optionally, a concave portion is disposed on a lower surface of the cover, and a protruding portion is disposed on an upper surface of the box; and when the cover covers the box, the protruding portion is inlaid in the concave portion, such that the internal space of the box is sealed by the cover.

[0013] Optionally, the ice bag further comprises:

[0014] Straps which are disposed on two sides of the ice bag and are of a double-layer structure.

[0015] Optionally, when the ice bag is placed flat, a non-zero second angle is formed between the straps and a vertical direction.

[0016] Optionally, the second angle is 3° .

[0017] Optionally, the cross-section of the box is hexagonal.

[0018] Optionally, the cross-section of the box is constituted by an antiparallelogram and a rectangle or a square.

[0019] Optionally, the cross-section of the box is octagonal.

[0020] Compared with the prior art, the application has the following beneficial effects:

[0021] 1. The zipper can be pulled open more easily without compromising the thermal-insulation property.

[0022] 2. The fitness of cotton and the zipper is improved, thus improving the thermal-insulation property.

[0023] 3. The firmness of the straps and the durability of the product are improved, and the service life of the product is prolonged.

[0024] 4. The space is effectively used, and the ice bag is smaller under the same capacity.

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] To better clarify other features, objectives and beneficial effects of the application, accompanying drawings constituting one part of the application are used to provide a further understanding of the application. The accompanying drawings and description of illustrative embodiments of the application are used to explain the application, and are not intended to improperly limit the application. Wherein,

[0026] FIG. 1 is an overall structural view of an ice bag according to one embodiment of the application;

[0027] FIG. 2 is a schematic diagram of a zipper pull according to one embodiment of the application;

[0028] FIG. 3 is a sectional view of an ice bag according to one embodiment of the application;

[0029] FIG. 4 is a sectional view of an ice bag according to one embodiment of the application;

[0030] FIG. 5 is a schematic diagram of straps according to one embodiment of the application;

[0031] FIG. 6 is a cross-sectional view of an ice bag according to one embodiment of the application.

DETAILED DESCRIPTION OF THE INVENTION

[0032] To allow those skilled in the art to gain a better understanding of the application, the technical solutions of the embodiments of the application will be clearly and completely described in conjunction with the accompanying drawings in the embodiments of the application. Clearly, the embodiments described below are merely illustrative ones, and are not all possible ones of the application. All other embodiments obtained by those ordinarily skilled in the art

based on the following ones without creative labor should also fall within the protection scope of the application.

[0033] It should be noted that the terms “first” and “second” in the description, claims and accompanying drawings of the application are used to distinguish similar objects, and are not necessarily used to describe a specific sequence or a sequential order. It should be understood that such data may be exchanged under appropriate circumstances to facilitate the description of the embodiments of the application. In addition, the terms such as “comprise”, “provided with” and any transformations thereof are intended to indicate non-exclusive inclusion. For example, a process, method, system, product or device comprising a series of steps or units is not definitely limited to those steps or units listed, and may comprise other steps or units that are not listed, or other inherent steps or units of the process, method, product or device.

[0034] It should be noted that the embodiments and features in the embodiments of the application may be combined without conflicts. The application will be described in detail below with reference to the accompanying drawings and embodiments.

[0035] Referring to FIG. 1-FIG. 6, one embodiment of the application provides an ice bag, comprising a cover 1 and a box 2, wherein the box 2 and the cover 1 are connected through a zipper which is provided with a zipper pull 31, and a portion, in contact with fingers of users, of the zipper pull 31 is arc.

[0036] In one embodiment of the application, when the ice bag is placed flat, a non-zero first angle α is formed between at least a tail of the zipper 3 and a horizontal direction.

[0037] In one embodiment of the application, a protruding portion 11 is disposed on a lower surface of the cover 1, and a concave portion 21 is disposed on an upper surface of the box 2; and when the cover 1 covers the box 2, the protruding portion 11 is inlaid in the concave portion 21, such that the internal space of the box is sealed by the cover.

[0038] In one embodiment of the application, a concave portion 12 is disposed on a lower surface of the cover, and a protruding portion 22 is disposed on an upper surface of the box; and when the cover covers the box, the protruding portion 22 is inlaid in the concave portion 12, such that the internal space of the box is sealed by the cover.

[0039] In one embodiment of the application, the ice bag further comprises straps 4, which are disposed on two sides of the ice bag and are of a double-layer structure.

[0040] In one embodiment of the application, when the ice bag is placed flat, a non-zero second angle β is formed between the straps and a vertical direction.

[0041] In one embodiment of the application, the second angle β is 3° .

[0042] In one embodiment of the application, the cross-section of the box is hexagonal.

[0043] In one embodiment of the application, the cross-section of the box is constituted by an antiparallelogram 51 and a rectangle or a square 52.

[0044] In one embodiment of the application, the cross-section of the box is octagonal.

[0045] In the application, the T-shaped zipper pull 31 is improved by designing the portion, in contact with the fingers of users, of the zipper pull to be arc, such that the zipper pull can better fit the fingers. The contact area between the zipper pull and the fingers is enlarged to reduce the pressure generated when the zipper pull is pulled with

fingers, thus relieving the sense of strenuousness when the zipper pull is pulled by users. In addition, the ice bag is generally erected on the ground when to be opened by users, then the users bow or squat to pull the ice bag open; generally, under this circumstance, the direction of force applied by hand to pull the ice bag open is not located on the same horizontal line as the direction of the zipper of the ice bag, such that only a few part of force applied by users can be effectively used to open or close the zipper. In the application, the angle α is designed at the head and tail of the zipper, such that when opening or closing the ice bag, users can apply force in a direction approximately the same as the direction of the zipper, such that users can apply force to the zipper more effectively. However, the application is not limited to this design, and an angle may be formed between the whole zipper and the horizontal direction.

[0046] The shape of cotton in the ice bag is optimized. Step surfaces similar to mortises and tenons are designed on cotton at the top and the cotton on a bag body and are matched with each other. When the cover covers the ice bag, the female surface of the cover fits the male surface of the bag body, such that the heat loss is effectively reduced, and the seam between the cover and the bag body is reduced. In the application, there is no limitation in the aspect whether the cotton on the top or the cotton on the bag body is female surface. The cotton at the top may be designed into a male surface (convex portion), and the surface of the bag body may be designed into a female surface (concave portion), vice versa. In addition, the design that the convex portion fits the concave portion does not definitely means that the convex portion is completely inserted into the concave portion (as shown in FIG. 4), and the convex portion may be partially inserted into the concave portion (as shown in FIG. 3).

[0047] In the application, through the design of the double-layer straps, the force-bearing capacity of the straps is greatly improved. In addition, a 3° angle is formed between the front and back straps and the vertical surface, and the straps are in the direction in which force is applied to handles, such that the attractiveness of the ice bag is guaranteed, and the bar-tack can act more effectively in the force-applying direction.

[0048] In the application, the overall shape of the ice bag is changed to be hexagonal, the front half of the ice bag is designed to be in the shape of a trapezoid (isosceles trapezoid or not), such that the internal spaces of the ice bags can be exactly staggered like ring-pull cans when the ice bags are stacked, thus improving the utilization rate of the internal space of the ice bag.

[0049] Compared with the prior art, the application has the following beneficial effects:

[0050] Compared with the prior art, the application has the following beneficial effects:

[0051] 1. The zipper can be pulled open more easily without compromising the thermal-insulation property.

[0052] 2. The fitness of cotton and the zipper is improved, thus improving the thermal-insulation property.

[0053] 3. The firmness of the straps and the durability of the ice bag are improved, and the service life of the ice bag is prolonged.

[0054] 4. The space is effectively used, and the ice bag is smaller under the same capacity.

[0055] The above description is merely used to explain preferred embodiments of the application, and is not

intended to limit the application. Those skilled in the art can make various modifications and transformations to the application. Any amendments, equivalent substitutions, and improvements made based on the spirit and principle of the application should fall within the protection scope of the application.

What is claimed is:

1. An ice bag, comprising a box and a cover, wherein: the box and the cover are connected through a zipper which is provided with a zipper pull, and a portion, in contact with fingers of users, of the zipper pull is arc; a protruding portion is disposed on a lower surface of the cover, and a concave portion is disposed on an upper surface of the box; and when the cover covers the box, the protruding portion is inlaid in the concave portion, such that an internal space of the box is sealed by the cover.
2. The ice bag according to claim 1, wherein when the ice bag is placed flat, a non-zero first angle is formed between at least a tail of the zipper and a horizontal direction.

3. The ice bag according to claim 2, wherein the ice bag further comprises:

straps which are disposed on two sides of the ice bag and are of a double-layer structure.

4. The ice bag according to claim 3, wherein when the ice bag is placed flat, a non-zero second angle is formed between the straps and a vertical direction.

5. The ice bag according to claim 4, wherein the second angle is 3°.

6. The ice bag according to claim 5, wherein a cross-section of the box is hexagonal.

7. The ice bag according to claim 6, wherein the cross-section of the box is constituted by an antiparallelogram and a rectangle or a square.

8. The ice bag according to claim 7, wherein the cross-section of the box is octagonal.

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